



Laboratory work №3

Databases

Tolepказы Galamat

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1. Retrieve all airline names in uppercase.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'airline' table selected under 'Tables (10)'. The SQL editor in the center contains the following query:

```
1 SELECT UPPER(airline_name) AS airli
2 FROM airline;
```

The 'Execute script' button (F5) is highlighted. The Data Output pane at the bottom shows the results of the query:

airline_name
IPC
PDN
KLE
KHS

A green status bar at the bottom indicates: 'Successfully run. Total query runtime: 63 msec. 50 rows affected.'

2. Replace any occurrence of the word "Air" in airline names with "Aero".

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'airline' table selected under 'Tables (10)'. The SQL editor in the center contains the following query:

```
1 SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
2 FROM airline;
```

The 'Execute script' button (F5) is highlighted. The Data Output pane at the bottom shows the results of the query:

airline_name
IPC
PDN
KLE
KHS

A green status bar at the bottom indicates: 'Successfully run. Total query runtime: 68 msec. 50 rows affected.'

3. Find all flight numbers that coordinates with both airline 1 and airline 2.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema, with 'Tables (10)' expanded. The SQL editor in the center contains the following query:

```
1 SELECT flight_no
2 FROM flights
3 WHERE airline_id IN (1, 2);
```

The Data Output pane at the bottom shows the results of the query:

flight_no
BR-PE
MZ-G
AU-NT
FR-K

A status message at the bottom right indicates: "Successfully run. Total query runtime: 70 msec. 46 rows affected." The status bar shows "Total rows: 46" and "Query complete 00:00:00.070".

4. Retrieve airports that contain the word "Reginal" and "Air" in their names.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema, with 'Tables (10)' expanded. The SQL editor in the center contains the following query:

```
1 SELECT airport_name
2 FROM airport
3 WHERE airport_name ILIKE '%Reginal%'
4 AND airport_name ILIKE '%Air%';
```

The Data Output pane at the bottom shows the results of the query:

airport_name

A status message at the bottom right indicates: "Successfully run. Total query runtime: 68 msec. 0 rows affected." The status bar shows "Total rows: 0" and "Query complete 00:00:00.068".

5. Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'passengers' table selected. The SQL query editor contains the following query:

```
SELECT
    first_name || ' ' || last_name AS full_name,
    TO_CHAR(date_of_birth, 'FMonth DD, YYYY') AS birth_date
FROM passengers;
```

The Data Output pane shows the results of the query, displaying columns 'full_name' and 'birth_date'. The results are as follows:

	full_name	birth_date
1	Hilde Iris	January 03, 2000
2	Arvy Sparsholt	June 09, 1974
3	Reinald Pococke	June 07, 1982
4	Con Borrel	October 17, 1986

The status bar at the bottom indicates: "Successfully run. Total query runtime: 70 msec. 200 rows affected." The status bar also shows "CRLF" and "Ln 2, Col 48".

6. Find flight numbers that have been delayed based on the actual arrival time.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'flights' table selected. The SQL query editor contains the following query:

```
SELECT flight_no
FROM flights
WHERE actual_arrival > scheduled_arrival;
```

The Data Output pane shows the results of the query, displaying the 'flight_no' column. The results are as follows:

	flight_no
1	US-CT
2	US-NM
3	RU-KR
4	US-AZ

The status bar at the bottom indicates: "Successfully run. Total query runtime: 73 msec. 509 rows affected." The status bar also shows "CRLF" and "Ln 3, Col 42".

7. Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema, with 'Tables (10)' expanded. The SQL editor in the center contains the following query:

```
SELECT
    flight_no,
    scheduled_arrival,
    actual_arrival
FROM flights
WHERE actual_arrival > scheduled_arrival;
```

The Data Output pane at the bottom displays the results of the query, showing 509 rows. The columns are flight_no, scheduled_arrival, and actual_arrival. The status bar indicates 'Successfully run. Total query runtime: 81 msec. 509 rows affected.'

flight_no	scheduled_arrival	actual_arrival
US-CT	2023-09-08	2023-11-07
US-NM	2023-09-17	2024-01-23
RU-KR	2023-03-18	2023-04-07
US-AZ	2023-04-08	

8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema, with 'Tables (10)' expanded. The SQL editor in the center contains the following query:

```
SELECT
    airline_name,
    airline_country,
    created_at
FROM airline
WHERE airline_country IN ('France', 'Portugal', 'Poland')
AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
```

The Data Output pane at the bottom displays the results of the query, showing 7 rows. The columns are airline_name, airline_country, and created_at. The status bar indicates 'Successfully run. Total query runtime: 75 msec. 7 rows affected.'

airline_name	airline_country	created_at
KHS	France	2023-11-18
IFH	Portugal	2024-01-09
KKG	Poland	2023-12-24
RUG	Portugal	

9. Find top3 overweighted baggage with more than 25kg.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'baggage' table selected. The SQL editor in the center contains the following query:

```
SELECT
  baggage_id,
  weight_in_kg,
  booking_id
FROM baggage
WHERE weight_in_kg > 25
ORDER BY weight_in_kg DESC
LIMIT 3;
```

The Data Output pane at the bottom shows the results of the query:

	baggage_id [PK] integer	weight_in_kg numeric (4,2)	booking_id integer
1	321	49.80	349
2	61	49.57	464
3	102	49.44	53

A status message at the bottom right indicates: "Successfully run. Total query runtime: 86 msec. 3 rows affected."

10. Find the youngest passengers' full name.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'passengers' table selected. The SQL editor in the center contains the following query:

```
SELECT
  CONCAT(first_name, ' ', last_name) AS full_name,
  date_of_birth
FROM passengers
WHERE date_of_birth = (
  SELECT MAX(date_of_birth)
  FROM passengers
);
```

The Data Output pane at the bottom shows the results of the query:

	full_name text	date_of_birth date
1	Bernie Mich...	2009-11-03

A status message at the bottom right indicates: "Successfully run. Total query runtime: 100 msec. 1 rows affected."

11. Find the cheapest booking price on each booking platform.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```
SELECT
  booking_platform,
  MIN(price) AS cheapest_price
FROM booking
GROUP BY booking_platform
ORDER BY booking_platform;
```

The Data Output tab shows the results of the query, displaying the booking platform and the cheapest price for each. The results are as follows:

booking_platform	cheapest_price
Abernathy-Metz	7439.50
Abshire, Bernier and Terry	9977.57
Adams-Greenholt	1017.87
Adams-Mertz	90

The status bar indicates: Total rows: 494, Query complete 00:00:00.074. A message box at the bottom right states: "Successfully run. Total query runtime: 74 msec. 494 rows affected."

12. Return airlines whose airline_code contains a digit.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```
SELECT
  airline_id,
  airline_code,
  airline_name
FROM airline
WHERE airline_code ~ '[0-9]';
```

The Data Output tab shows the results of the query, displaying the airline_id, airline_code, and airline_name for the single row that matches the criteria. The results are as follows:

airline_id	airline_code	airline_name
22	2AK6	HGZ

The status bar indicates: Total rows: 1, Query complete 00:00:00.074. A message box at the bottom right states: "Successfully run. Total query runtime: 74 msec. 1 rows affected."

13. List the top5 most recently created airlines.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema, including tables like 'airline', 'airport', 'baggage', etc. The main query editor displays the following SQL query:

```
SELECT
  airline_id,
  airline_name,
  created_at
FROM airline
ORDER BY created_at DESC
LIMIT 5;
```

The Data Output pane shows the results of the query:

	airline_id [PK] integer	airline_name character varying (50)	created_at date
1	9	NQX	2024-03-16
2	40	CII	2024-03-15
3	22	HGZ	2024-03-13
4	24	RUG	2024-03-13
5	15	RUG	2024-03-13

A green status bar at the bottom indicates: "Successfully run. Total query runtime: 75 msec. 5 rows affected."

14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema. The main query editor displays the following SQL query:

```
SELECT *
FROM baggage_check
WHERE booking_id BETWEEN 200 AND 300
AND check_result <> 'Checked';
```

The Data Output pane shows the results of the query:

	baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	7	Not checked	2023-05-04	2023-04-29	226	44
2	12	Not checked	2023-10-02	2023-06-22	288	133
3	22	Not checked	2023-06-14	2023-11-10	246	101
4	23	Not checked	2023-06-14	2023-11-10	246	101

A green status bar at the bottom indicates: "Successfully run. Total query runtime: 81 msec. 56 rows affected."

15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'baggage' table selected under 'Tables (10)'. The main pane shows a SQL query executed against the 'postgres/postgres@PostgreSQL 18' database. The query filters for baggage checks where the month of 'update_at' matches the month of 'created_at', but 'update_at' is earlier than 'created_at'. The results pane shows 18 rows, with the first four rows displayed. A status message at the bottom indicates the query was successfully run with a total runtime of 78 msec, affecting 18 rows.

```
SELECT *
FROM baggage_check
WHERE EXTRACT(MONTH FROM update_at) =
      EXTRACT(MONTH FROM created_at)
AND EXTRACT(YEAR FROM update_at) =
      EXTRACT(YEAR FROM created_at)
AND update_at < created_at;
```

	baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	34	Checked	2023-04-25	2023-04-08	28	57
2	86	Checked	2023-12-25	2023-12-07	312	22
3	104	Checked	2024-03-17	2024-03-14	41	83
4	109	Not checked				

Successfully run. Total query runtime: 78 msec. 18 rows affected.